

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Mark s
Q. 1	Objective type questions. (Compulsory Question)		12
1	What happens to the resistance of a metal conductor when its temperature increases?	CO2	1
	A) It decreases B) It remains the same C) It increases D) It becomes zero		
2	What is the unit of potential difference?	CO1	1
	A) Ampere B) Watt C) Ohm D) Volt		
3	Instrument used to measure electric current:	CO1	1
	A) Ammeter B) Voltmeter C) Wattmeter D) Energy meter		
4	MCB protects against:	CO2	1
	A) Voltage dips B) Earth faults C) Overcurrent and short circuit D) Frequency fluctuation		
5	Which method is commonly used for 3-phase power measurement?	CO2	1
	A) Single wattmeter method B) Three ammeters C) Two wattmeter method D) Power triangle method		
6	Ripple factor (r) of a half-wave rectifier is:	CO2	1
	A) 0.121 B) 1.21 C) 0.482 D) 0.906		

7	Which filter uses only a capacitor connected in parallel with the load?	CO3	1
	A) Choke input filter B) π filter C) Capacitor filter D) RC filter		
8	Which diode emits light when forward biased?	CO3	1
	A) Zener diode B) Tunnel diode C) LED D) Photodiode		
9	Which number system uses only digits 0 and 1?	CO1	1
	A) Decimal B) Octal C) Hexadecimal D) Binary		
10	Which gate gives a HIGH output only when all inputs are HIGH?	CO2	1
	A) AND B) NOT C) NOR D) OR		
11	The binary equivalent of the decimal number 13 is	CO1	1
	A) 1101 B) 1110 C) 1011 D) 1001		
12	A cut in Voltage for Germanium Diode is....	CO2	1
	A) 0.7 V B) 7 V C) 1.3 V D) 0.3 V		
Q. 2	Solve the following.		12
A)	Explain the concept of potential difference and electric current and derive Ohm's Law.	CO1	6
B)	Classify energy resources into conventional and non-conventional types and explain any three non-conventional energy resources. https://www.pyqonline.com	CO2	6
Q.3	Solve the following. s		12
A)	Explain how diode working under reverse bias.	CO3	6
B)	Explain following term: i. Generation of carrier ii. Recombination of Carrier iii. Diffusion and Diffusion Current	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain with a neat diagram the structure of an electrical power system, including generation, transmission, distribution, and utilization.	CO1	6
B)	Explain the differences between single-phase and three-phase power supply.	CO3	6

C)	Explain the working principle and applications of a Miniature Circuit Breaker (MCB) and a Motor Protection Circuit Breaker (MPCB).	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain half wave rectifier.	CO2	6
B)	Describe construction and working of LED.	CO2	6
C)	Explain Characteristics of Full wave rectifier.	CO2	6
Q. 6	Solve Any Two of the following.		12
A)	i. Convert $(D7BA9.E8C)_{16}$ into $(\quad)_8$ ii. Convert $(97.27)_{10}$ into $(\quad)_2$ iii. Convert $(110101.011)_2$ into $(\quad)_{10}$	CO1	6
B)	Explain universal gates.	CO1	6
C)	Define a transistor. Explain its construction and working principle.	CO2	6
	*** End ***		